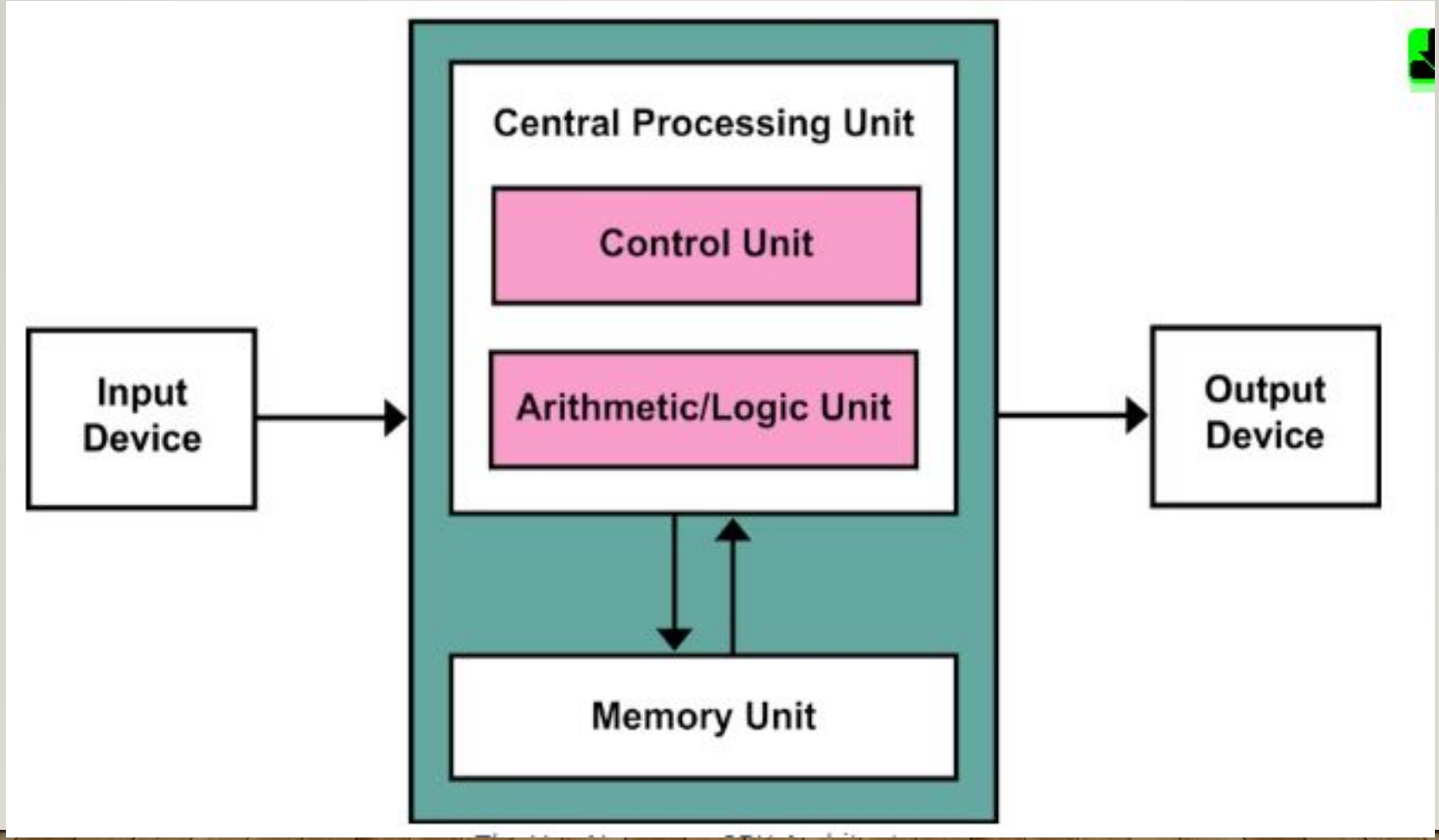
Modern CPUs

Understanding Modern CPU Architecture:



- The Central Process Unit is usually referred to as "The Brain" of the computer. The CPU is at the heart of all data interpretation and processing operations in a computer.

- The CPU contains three major components: memory, control unit, and the ALU(arithmetic and logic unit).

- The modern CPU architecture is implemented on ICs(Integrated Circuits), with one or two single metal-oxide-semiconductor IC chips.

- Memory: A memory component for storing data and instructions
- Central Processing Unit: for decoding and executing instructions
- Inputs/Outputs: and a set of inputs and output interfaces

A microprocessor is an integrated circuit containing a CPU, responsible for executing instructions in a computer. The CPU is the core processing unit, handling computations and control functions.

- A chip's architecture determines where its parts are located and connected, how it connects with other parts of the computer, and much more.
- A processor's architecture determines how many transistors it has, and therefore the processor's power

- A processor includes many other features that affect its performance. For example, a processor's performance is affected by the number of bits of data it can process at any one time.
- In 2003 American Micro Devices (AMD) released a new generation of desktop PC processor that can handle 64 bits of data.

- A 64-bit processor refers to a microprocessor that can process data and instructions in chunks of 64 bits.

Look Inside the CPU:

- Architecture:

  Determines
- Location Of CPU.
- Bit size
- determines how many transistors it has

Intel:

- Intel is historically the leading provider of chips for PCs.
- 1971, Intel invented the microprocessor the so-called computer on a chip—with the 4004 model.
- IBM pc powered by intel 8086.
- Current processors:

a) Xeon.
b) Core i7, i5, i3
c) Pentium.

Microcomputer Processors:

- Advanced Micro Devices(AMD)
- Main competitor to intel.
- Originally produced budget products
- Current products outperform Intel
- Current processors

Opteron.
FX.
Athlon.

Freescale:

- A Subsidiary of Motorola.
- Co-Developed the Apple G4 PowerPC
- Currently focus on linux Market

800 MHz Motorola PowerPC 7450 processor on the CPU module of a Power Mac G4

IBM:

- Historically manufactured mainframes.
- Partnered with Apple to develop G5
  First Consumer 64 bit

Comparing Processors:

- Speed of processor.
- Size of cache.
- Number of registers.
- Bit size.
- Speed of side bus.

Advanced Processor Topics:

- **RISC Processors**

 Reduced Instruction Set Computing.
 Smaller instruction sets.
 May process data faster.
 PowerPC and G5.

- **CISC Processors**

  Complex Instruction Sets Computing.
 Large instruction sets.
 More Applications.
 IBM Compatible.

Parallel Processing

- Multiple processors in a system.
- Symmetric Multiple Processing
  - Number of processors is a power of 2.

- Massively Parallel Processing.
  - Thousands of processors.

Standard computer port:

- Keyboard and mouse ports.
- USB ports.
- Parallel ports.
- Network ports.
- Modem ports.
- Audio ports.
- Serial ports.
- Video ports.

Extending The Processors Power

- **SCSI**

- Small Computer System Interface.
- Supports of dozens of devices.
- External devices daisy chain.
- Fast hard drives and CD-ROMs.

USB

- Universal Serial Bus.
- Most popular external.
- Supports up to 127 devices.
- Hot swappable.

FireWire:

- IEEE 1884
- Cameras and video equipment.
- Hot swappable.
- Port is very expensive.

Expansion slots and boards:

- Allows users to configure the machine.
- Slots allow the addition new devices
- Devices are stored on cards.
- Computer must be off before inserting.

Plug and play:

- New hardware detected automatically .
- Prompts to install drives.
- Non-technical users can install devices.

Storing Data

Storage : The purpose of a storage device is to hold data—even when the computer is turned off—so the data can be used whenever it is needed. Storage involves two processes:

- Writing, or recording, the data so it can be found later for use.
- Reading the stored data, then transferring it into the computer's memory.

The physical materials on which data is stored are called storage media. The hardware components that write data to, and read data from, storage media are called storage devices.

For example, a diskette is a storage medium (medium is the singular form of the word media); a diskette drive is a storage device.

- **Categorizing Storage Devices:**

The two main categories of storage technology used today

- **Magnetic Storage.**
- **Optical Storage.**

A third category of storage—solid-state storage—is increasingly being used in computer systems, but is more commonly found in devices such as digital cameras and media players.

Magnetic Storage Devices

- Most common form of storage.
- Hard drives, floppy drives tape.
- All magnetic drives work the same.
- diskette drives, hard disk drives, high-capacity floppy disk drives, and rape drives use similar techniques for writing and reading data.
- The surfaces of diskettes, hard disks, high-capacity floppy disks, and magnetic tape are coated with a magnetically sensitive material, such as iron oxide, that reacts to a magnetic field.

Floppy Disk
Hard Disk
Tape

- Data storage and retrieval

✔ Media is covered with iron oxide.

✔ Read/Write head is a magnet.

✔ Magnet writes charges on the media.

- Positive charge is a 1
- Negative charge is 0

✔ Magnet reads charges

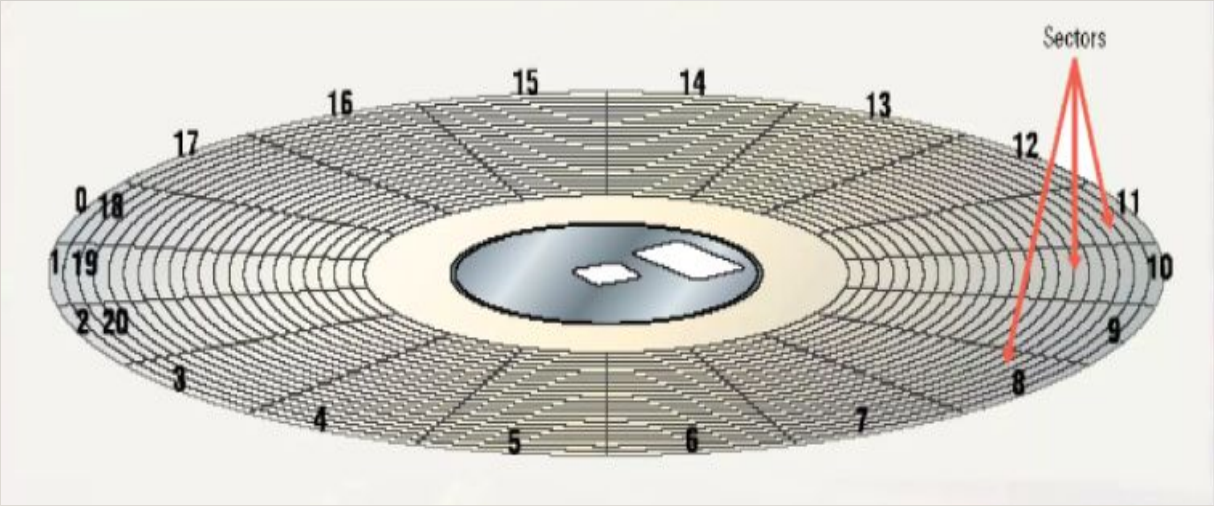
✔ Drive converts charges into binary

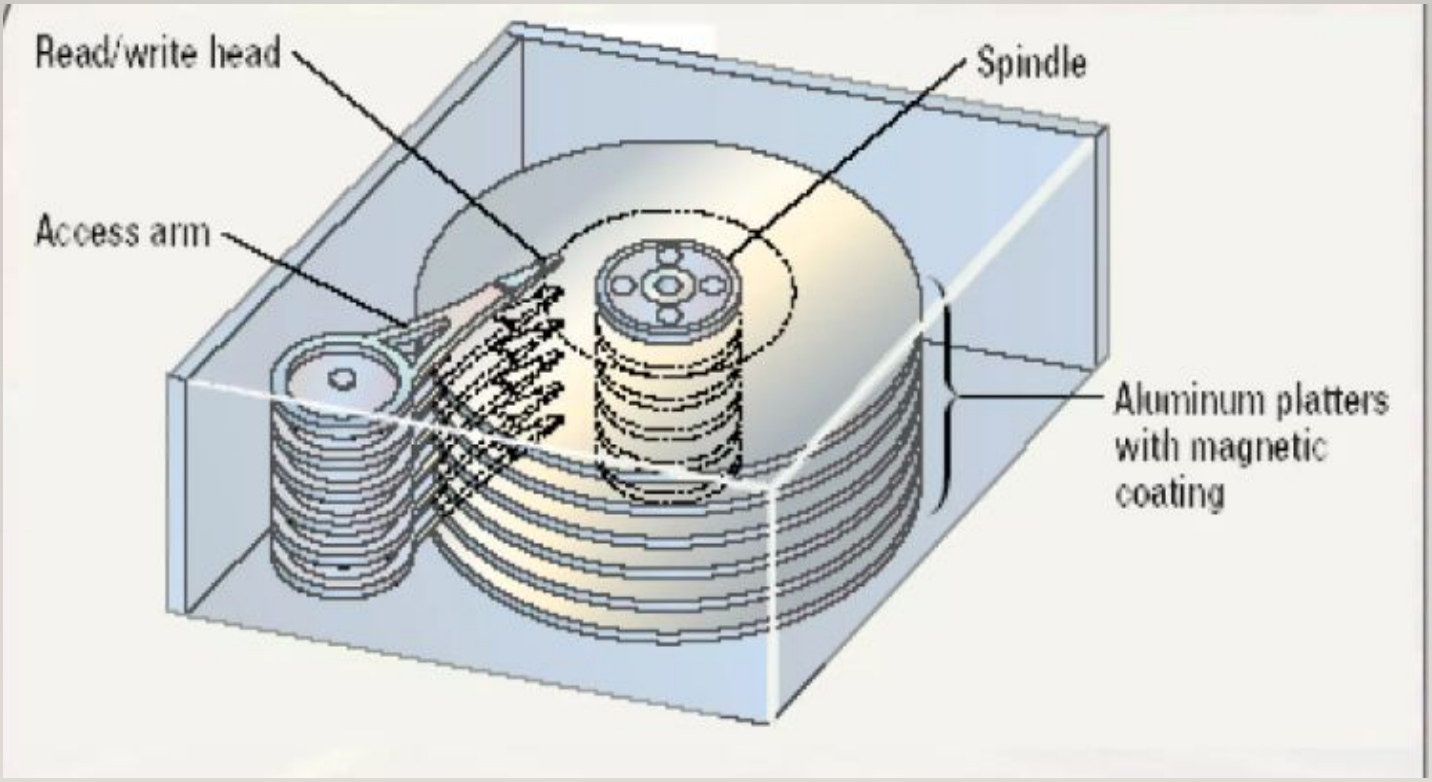
✔ Magnetic storage devices work by polarizing tiny pieces of iron on the magnetic medium. Read/write heads contain electromagnets that create magnetic charges on the medium.

Data Organization

- Disks must be formatted before use.
- Format draws tracks on the Disk
- Tracks is divided into sectors
- Amount a data drive can read.

Tracks and Sectors




Read/write head
Spindle
Access arm
Aluminum platters
with magnetic
coating

MAGNETIC

PROS	CONS
• Available with **large capacities** and **cheap to buy**	• Generally not **durable** (disks can break easily)
• Usually **reliable**, although failures do occur	• **Slower read/write** times compared to SSDs

Optical Storage Devices:

- Thee most popular alternatives to magnetic storage systems are optical systems, including CD-ROM, DVD-ROM, and their variants.
- These devices fall into the category of optical storage because they store data on a reflective surface so it can be read by a beam of laser light.

DVD-ROM

- Digital Video Disk
- Use both sides of the disk
- Capacities can reach 18 GB
- DVD players can read CDs

OPTICAL

~700 MB ~5 GB ~25 GB

- Examples include CDs, DVDs, and Blu-Ray
- They use different reflections of light to **read** data, and use a laser to **write** data

- The information is held on the disc, but you need a **disc drive** to access it

CD Recordable (CD-R)

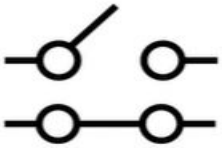
- Create a data or audio CD
- Data cannot be changed
- Can continue adding until full

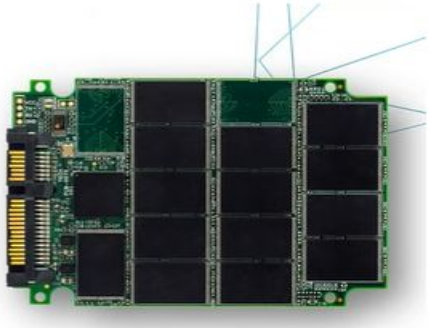
PROS	CONS
• **Reliable**	• Per unit, have a smaller **capacity** than alternative
• **Good portability** – easy to pack and carry	• **Slower read/write speed** compared to alternatives
	• Need a **disc drive** to use

Solid-state storage:

- Solid-state storage devices store data on memory circuits rather than disks or tapes. They store data electronically, not in a magnetic or optical form.
- Solid-state storage devices can use either volatile or nonvolatile memory: Examples of such devices include flash memory. smart cards, and solid-state disks.

SOLID STATE





- Have no moving parts
- Electronic circuits are used to store the data
- Examples include SSDs and **flash memory** devices like USB memory sticks and memory cards (like SD cards)

SOLID STATE

PROS	CONS
• Generally **reliable**	• **Higher cost** per GB
• Can be very **durable**	• Sometimes have a **limited number of writes** (leading to reliability problems long term)
• Can be **portable**	
• Generally **faster read/write** times	

Computer Networks

Contents:

- **Introduction**
- **Types of network**
- **Network basic terminology**
- **Types of network architecture**
- **Workgroup**
- **Domain**

❖ **What ?**

✔ A group or system of interconnected people or things

❖ **Why?**

✔ Connection

✔ Communication

❖ **Where?**

✔ When we need to transfer anything.

Computer Networks

- A computer network is a set of devices connected by communication links.

- A node can be a computer, printer or any other device capable of sending or receiving data generated by other nodes on the network.

Computer Network

Networking

- Networking is a process of communication between two or more remote parties, that involves the connection of computers media and networking devices.

Advantages of Networking

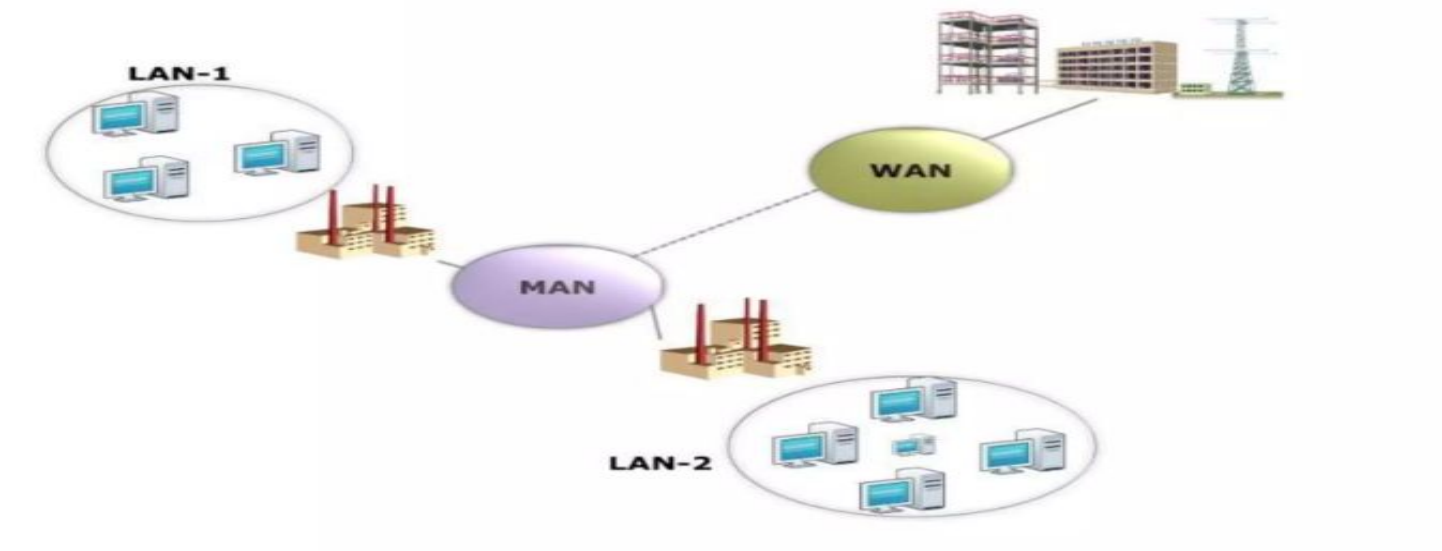
- **Easy Communication**
- File, data and information sharing
- Resource sharing(hardware)
- Increase storage capacity
- Reduce cost
- Save time

Network Components:

- **Devices:** Computers, servers, routers, switches, and other hardware connected to the network.
- **Cables and Connections:** Ethernet cables, fiber optic cables, and wireless connections (Wi-Fi).
- **Protocols:** Sets of rules governing communication between devices, such as TCP/IP, HTTP, and DNS.

Types of Network

- Physical structure-LAN-WAN-MAN



LAN (Local Area Network): A network confined to a small geographic area, such as a home, school, or office.

WAN (Wide Area Network): A network that spans a large geographic area, often connecting multiple LANs across cities or countries.

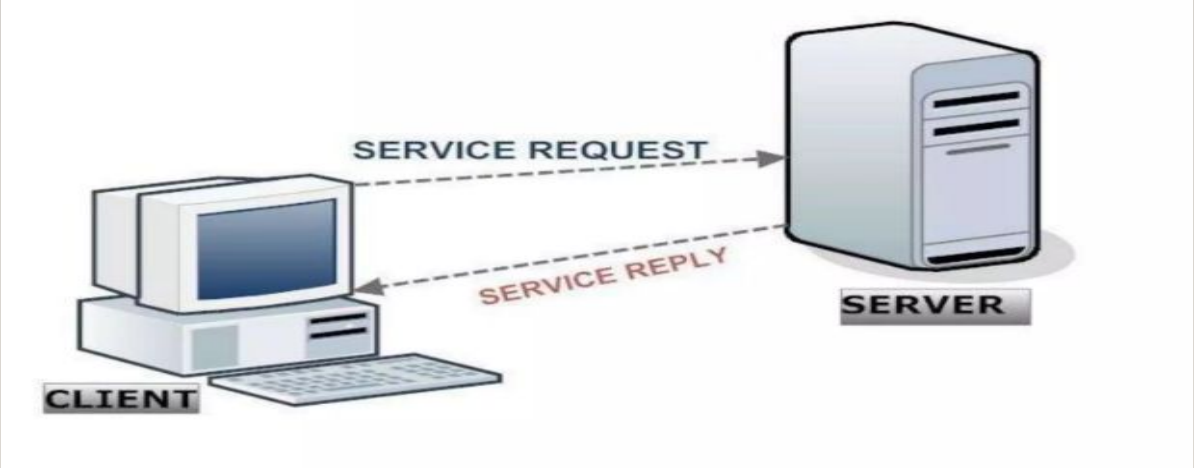
Metropolitan Area Network (MAN): A MAN is a network that spans a city or metropolitan area, providing connectivity between LANs within the same region.

Network Terms

- Host- Workstation- Server- Client-Node

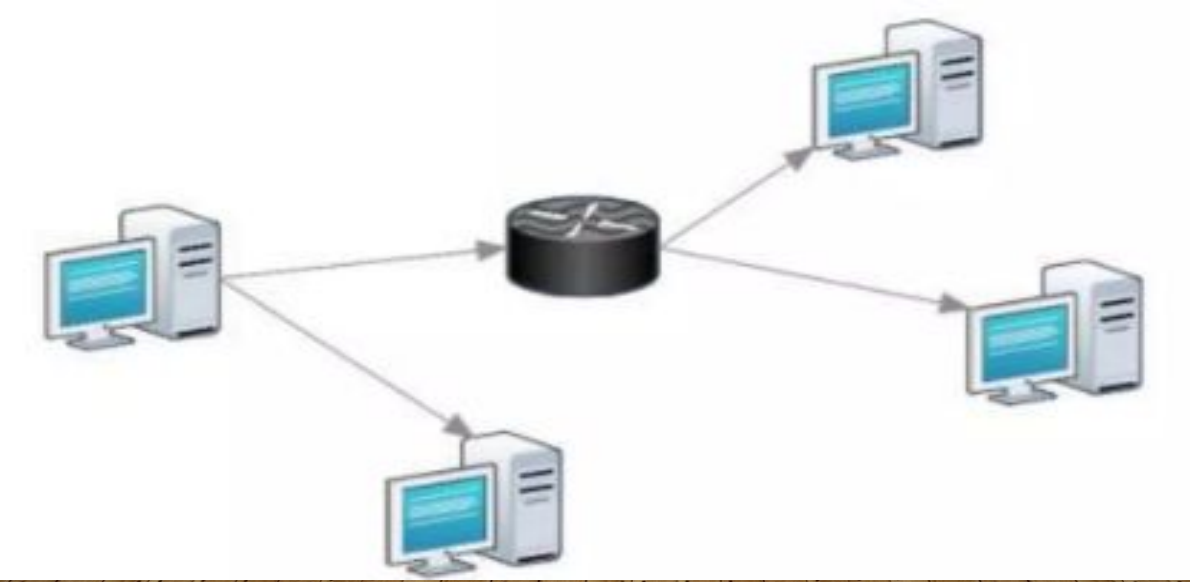
- **Host:** A network host is a computer or other device connected to a computer network.
- **Workstation:** A workstation is a computer dedicated to a user or group of users engaged in business or professional.

Client and Server



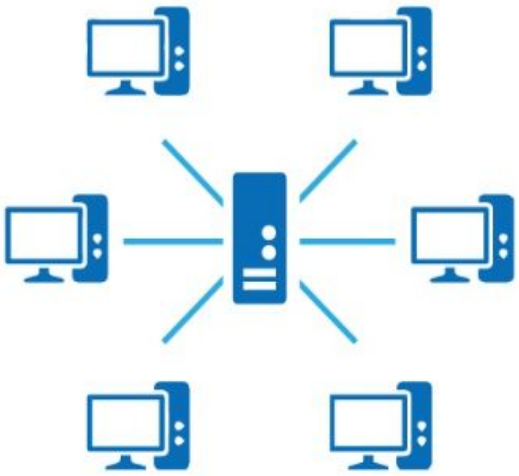
- For process to process communication we need Client Server Paradigm.
- A process om local host, called a client, needs Services from a process usually on the remote host called a server.
- A server is an a computer program that accepts and responds to requests made by another program, known as client.

- **Node:** A node can be a computer or some other device such as printer that can receive, create, store or send data along distributed network routes.

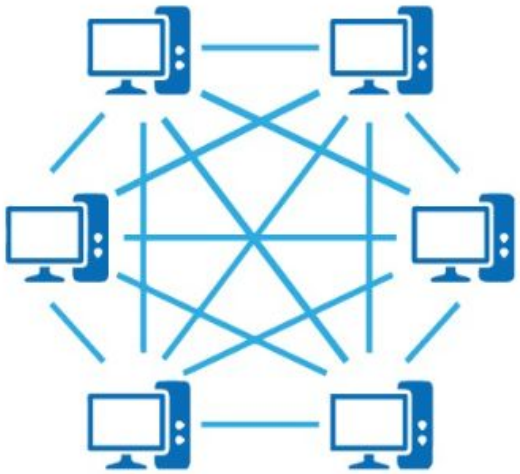


Types Of Network Architecture

- **Peer to Peer:** A peer to peer network is one in which two or more PCs share files and access to devices such as printers without requiring a separate server computer or server software.
- **Client-Server Network:** In client-server model, where client request for a service to a service and server replies according to client request.



A **Server** based Network

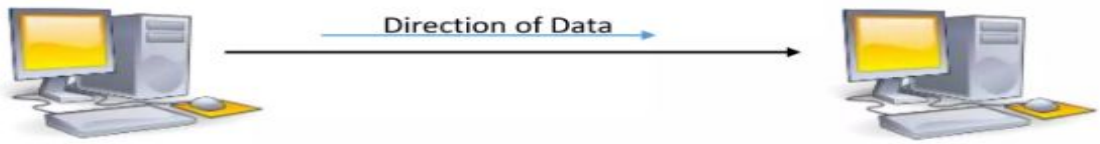


A **Peer-to-Peer** based Network

Data Flow

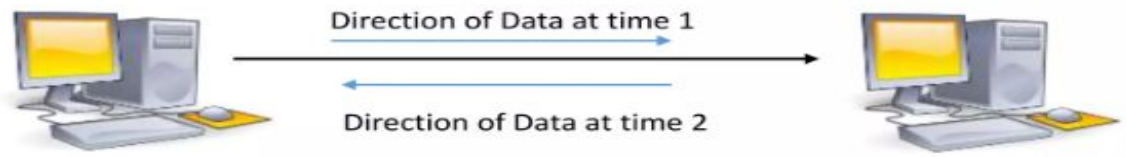
- Simplex- Half Duplex- Full Duplex

- In Simplex mode:
  - Communication is unidirectional,
  - Only one can transmit and other can receive.
  - Ex: keyboard and monitor



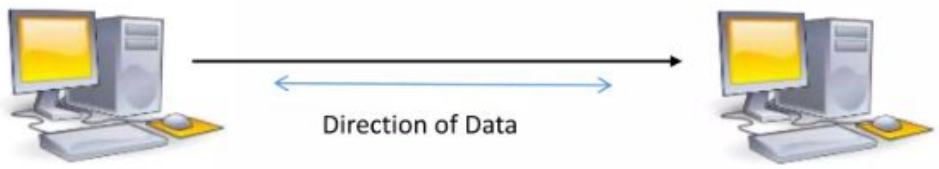
Simplex Communication Structure

- In Half Duplex mode:
  - Both can transmit and receive,
  - But not at a same time.
  - At a time, only one can send or receive.
  - Ex: Walkie-talkies



Half Duplex Communication Structure

- In Full Duplex mode:
  - Both station can transmit and receive simultaneously,
  - But at a same time.
  - Ex: Telephone Networks



Full Duplex Communication Structure

Workgroup Vs Domain

- **Workgroup:**
- A **workgroup** is a peer-to-peer network using common software. A workgroup allows all participating and connected systems to access shared resources such as files, system resources and printers.

Domain:

- In a **domain** is a subnetwork made up of a group of clients and servers under the control of one central security database.

Workgroup Vs Domain

Domain	Workgroup
• Computers on workplace networks are usually part of a domain.	• Computers on home networks are usually part of a workgroup.
• One or more computers are server.	• All computers are peers.
• For data security.	• No data security.

Computer Software

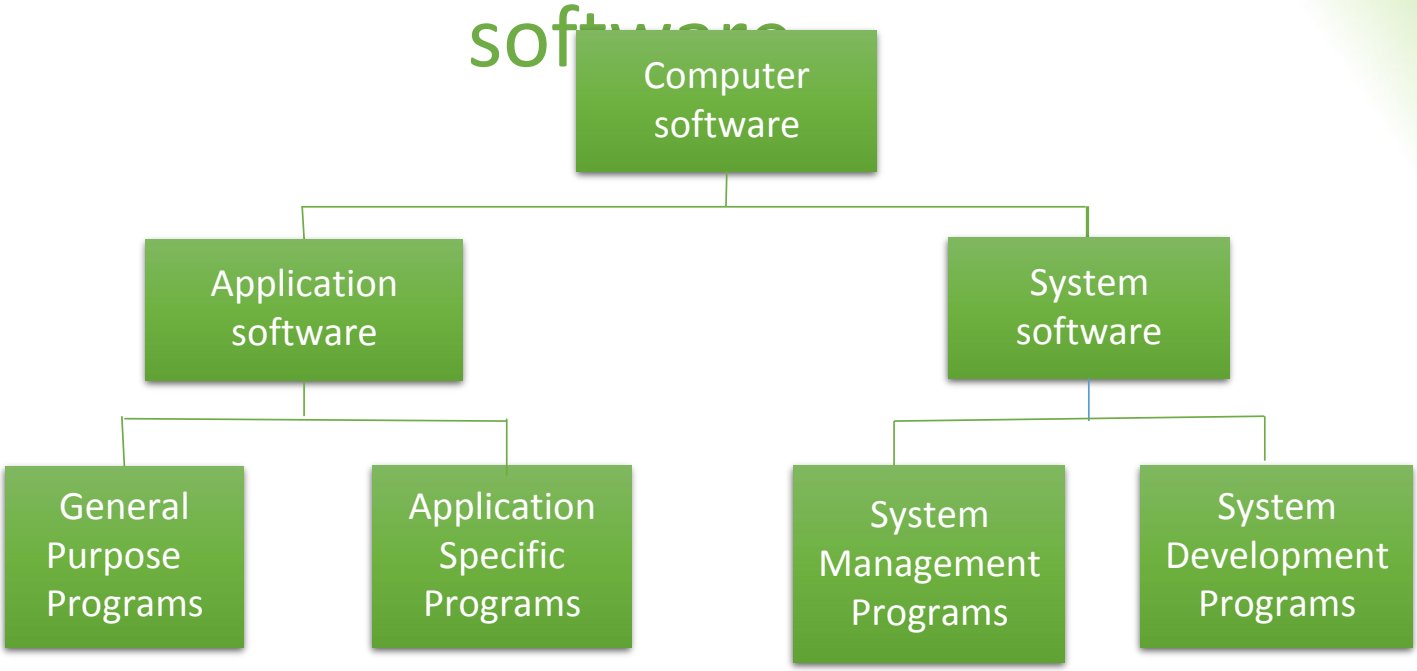
Computer Software

- **Software** is a set of instructions that drive a computer to perform specific tasks.
- These instructions tell the machine's physical components what to do.
- A set of instructions is often called a **program**.
- When a computer is using a particular program, it is
- said to be **running** or **executing** the program.
- The two most common types of programs are **system software** and **application software**.

Computer Software

- **System software:** helps the computer to carry out its basic operating tasks.
- **Application software:** helps the user carry out a variety of tasks.
- A Computer hardware is useless without software.

Types of software



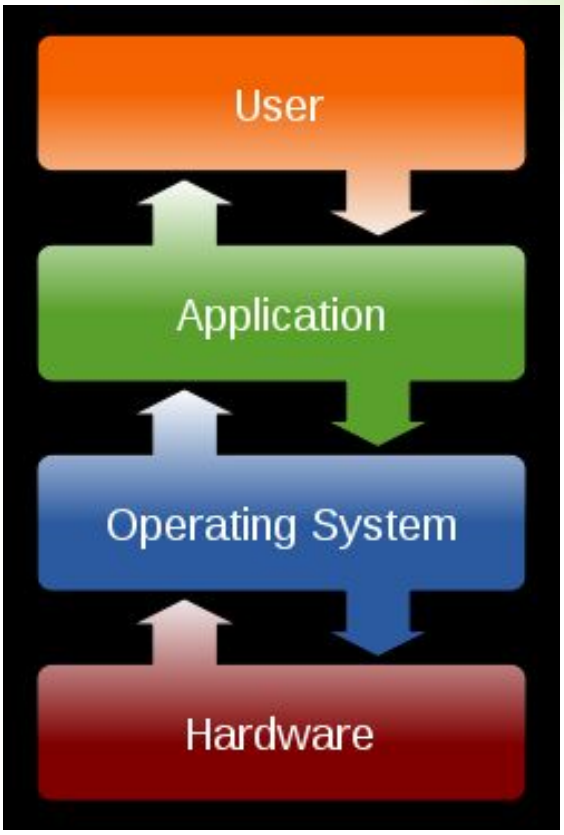
System Software

- **System Software**: computer software designed to operate the computer hardware, to provide basic functionality, and to provide a platform for running application software.
- It functionality is to bridge computer system hardware and application software.
- **Examples**
  - Device drivers
  - Utility programs
  - Operating systems
  - Compiler
  - Firmware
  - Networking software

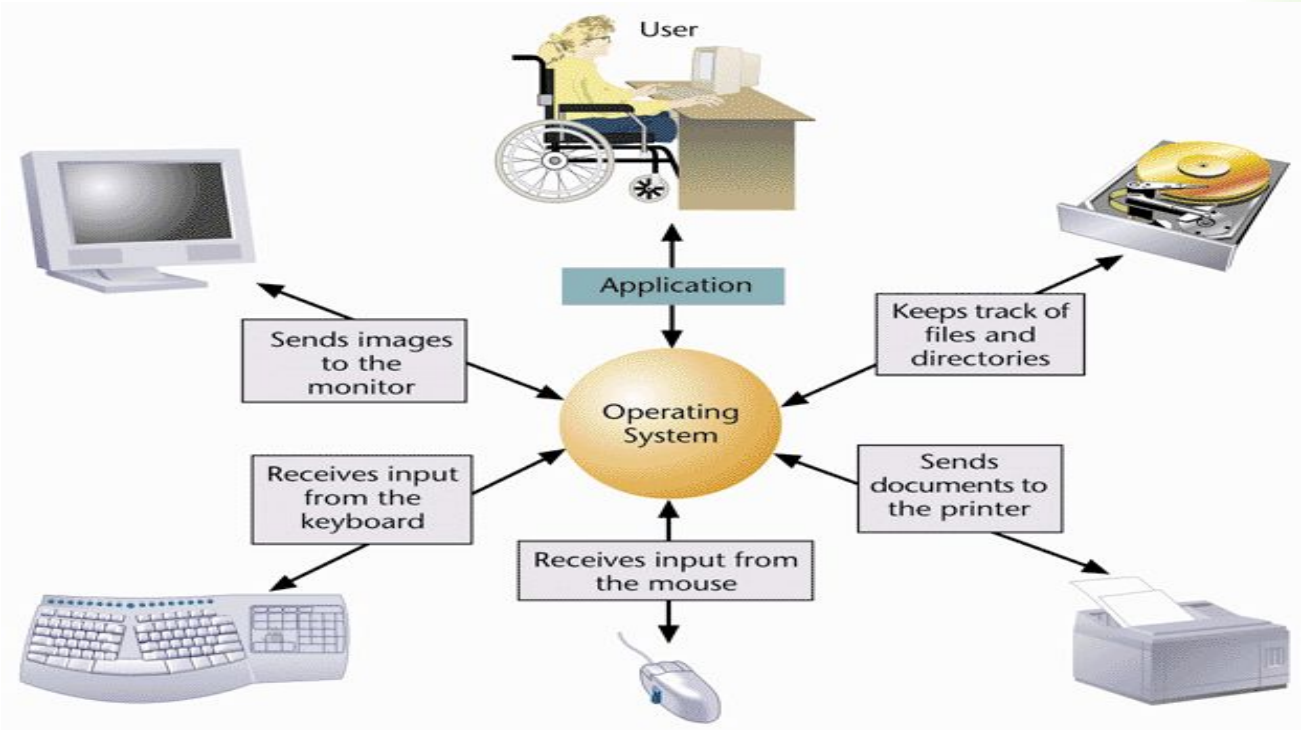
Operating System (OS)

- It is a software that **controls** the system's hardware and interacts with the user and application software.

- The operating system acts as an intermediary between application programs and the computer hardware.



Operating System (OS)



- Mediates between the application and computer, and control peripheral devices.

How OS works?

- When a computer is powered on, it loads the OS, normally from the disk drive into RAM.
- **Kernel:** portion of OS code that interact with the computer hardware.
- **Shell:** portion that interfaces with the applications and user using either the command line interface (CLI) or graphical user interface (GUI).
- CLI, user interacts directly with the system in text-based environment by entering commands on the keyboard at command prompt.
- GUI users interact with the system in an environment that uses graphical images, multimedia, and text. Its user friendly and requires less knowledge.

Functions of an OS

- Resource management: allocate computer resource such as CPU time, main memory, secondary storage, i/o devices for use.
- Data management: govern the input & output of data and their location storage and retrieval.
- Task management: monitors, schedules and control, prepares jobs for execution.
- Memory management
- File management
- Booting the computer
- Provides user interface
- Enforces protection policies
- Provides a variety of support services

Types of an operating system

Operating systems can be categorized according to availability, number of users, type of interface design and manufacturer:

According to manufacturer

- Microsoft window: XP, Vista, 2003 Server, windows7,8
- UNIX-Based
- Linux (Ubuntu)
- Macintosh OS

User Interface

- **User interface**
  - A function of the operating system that allows individuals to access and command the computer
- **Command-based user interface (CLI)**
  - A particular user interface that requires text commands be given to the computer to perform basic activities E.g., unix, DOS
- **Graphical user interface (GUI)**
  - A user interface that uses pictures (icons) and menus displayed on the screen to send commands to the computer system E.g. Windows, MAC OS

Types of an operating system

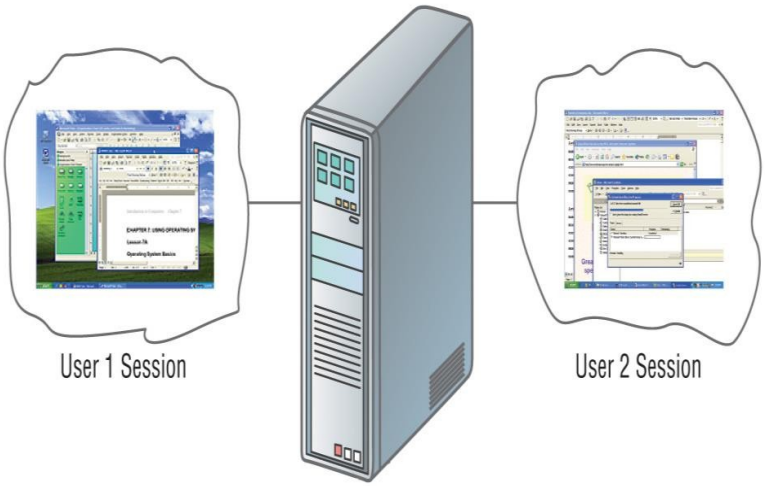
Single-User/Single-Tasking Operating Systems:

- Allows a single user to perform just one task at a time
- Take up little space on disk
- Run on inexpensive computers
- Examples include; MS-DOS and Palm OS for palm handheld computers.

Single-User/Multi-Tasking Operating Systems:

- Allows a single user to perform two or more functions at once.
- Commonly used on personal computers.
- Examples include; Microsoft Windows and MAC OS.

Types of an operating system



Multi-User/Multitasking Operating Systems:

- Allows multiple users to use programs that are simultaneously running on a single network server.
- Here, each user is given a user session on the server.
- UNIX, Linux are examples.
- Maintenance can be easy.
- Requires a powerful computer.

System Software

- **A utility program:** is a program used to enhance performance of the system. Utilities are used to manage disks, troubleshoot hardware problems….

- Some independent software developers offer utilities for

  sale separately. E.g. Norton Utilities by Symantec.

  For example

  - Symantec antivirus
  - Partition Magic
  - Disk defragmentation
  - Screen saver
  - Backup, etc…

System Software

- **Device driver:** A computer program that can establish communication because it contains information about the characteristics of your computer and of the device.
- Each peripheral device requires a device driver.
- Helps the computer communicate with that particular device.
- When we add a device to an existing computer, part of its installation includes adding its device driver to the configuration.

System Software

- **Compiler:** Software that translates a high-level language program into machine language.
- Input to the compiler is a source file (created by word processor or editor) containing the text of a high-level language program.
- If it is syntactically correct, compiler will save in an object file which is a machine language instructions for the same job.

Application Software

Application Software: Tells the computer how to accomplish specific tasks, such as word processing or drawing, for the user.

Most applications are purchased on diskette or CD-ROM.

They are installed by copying the programs from the diskettes/CD-ROM to the hard disk.

Payroll systems, Inventory Control, Manage student database

Categories of Applications Software

- Word processing software for creating text-based documents
- Spreadsheets for creating numeric-based documents such as budgets.
- Presentation programs for creating and presenting electronic slide shows.
- Graphics programs for designing illustrations or manipulating photographs, movies etc.
- Database management software for building and manipulating large sets of data such as names, addresses etc.

Word Processors:

It is a tool that helps user in creating, editing, and printing documents. Word processors will normally have the following capabilities built into them:

- Spell checking
- Standard layouts for normal documents
- Some characters appearing in bold print, italics, or underlined
- Center lines, make text line up on the left side of the
  paper, or the right side of the paper
- Save the document so it can be used again
- Print the document.

Examples: WordPerfect and Microsoft Word

Spreadsheets

The spreadsheet packages are designed to use numbers and formulas to do calculations with ease. Examples of spreadsheets include:

- Budgets
- Payrolls
- Grade Calculations
- Address Lists

The most commonly used spreadsheet programs are Microsoft Excel and Lotus 123.

Actions to prevent virus infection

- Forget opening unexpected e-mail attachments, even if they're from friends
- Get trained on your computer's anti-virus software and use it.
- Have multiple backups of important files. This lowers the chance that all are infected.
- Install security updates for your operating system and programs as soon as possible.
- Jump at the chance to learn more about your computer. This will help you spot viruses.

Security Software

- **Security software:** designed to protect computers from various forms of destructive software and unauthorized intrusions

- The terms <u>malicious software</u> and <u>malware</u> refer to any computer program designed to secretly enter a computer, gain unauthorized access to data, or disrupt normal processing operations

What is computer virus?

- Computer virus refers to a program which damages computer systems and/or destroys or erases data files
- Once it’s running, it spreads by inserting copies of itself into other executable code or documents

Types of Computer Virus

- Computer Worm
- Trojan Horse
- Spyware
- Time Bomb
- Logical Bomb
- Micros Virus
- Script Virus

What does malware/virus

- Display irritating messages and pop-up ads.
- Delete or modify your data.
- Encrypt your data and demand ransom for the encryption key.
- Upload or download unwanted files.
- Log your keystrokes to steal your passwords and credit card numbers.
- Propagate malware and spam to everyone on your e-mail address book or your instant messaging buddy list.
- Disable your antivirus and firewall software.
- Block access to specific Web sites and redirect your browser to infected Web sites.
- Cause response time on your system to deteriorate.
- Allow hackers to remotely access data on your computer.
- Allow hackers to take remote control of your machine and turn it into a zombie.
- Link your computer to others in a botnet that can send millions of spam e-mails or wage Denial of Service attacks against Web sites.
- Cause network traffic jams.

Symptoms of computer virus

- Computer runs slower than usual
- Display unusual error messages
- New icons appear on desktop
- Programs disappear from the computer
- Reformat the hard disk
- Duplicate of files
- Computer restarts all the time

Typical things that some current Personal Computer (PC) viruses do

- Display a message
- Erase files
- Scramble data on a hard disk
- Cause erratic screen behavior
- Halt the PC
- Many viruses do nothing obvious at all except spread!

Actions to prevent virus infection

- Always update your anti-virus software at least weekly.
- Back up your important files and ensure that they can be restored.
- Change the computer's boot sequence to always start the PC from its hard drive
- Don't share Drive C: without a password and without read-only restrictions.
- Empty floppy drives of diskettes before turning on computers, especially laptops.

Sources of computer virus

- Fake games
- Infected software
- Opening infected files
- Contact with infected systems or devices
- Free or shareware software

Doing Business in the Online World

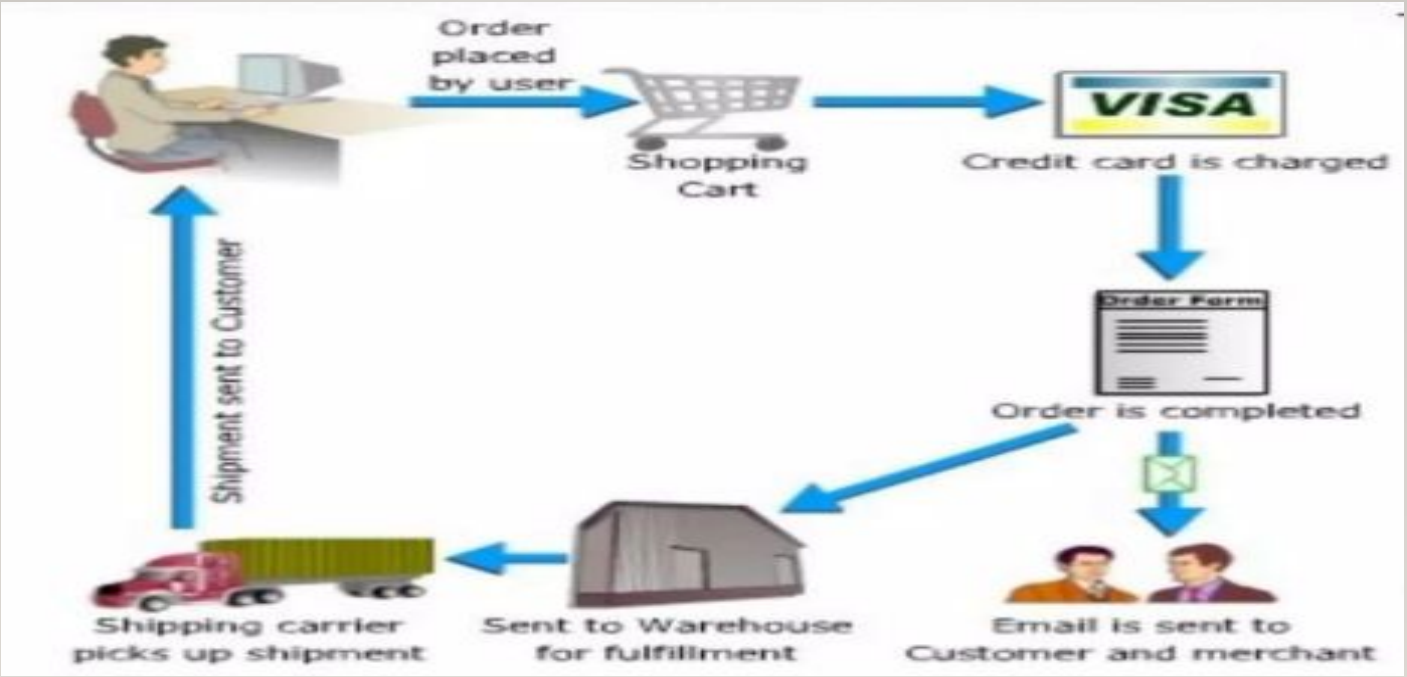
E-Commerce

- It is Commonly known as electronic marketing.
- It is consist of buying and selling goods and services over an electronic system such as the internet.
- E-commerce is the purchasing, selling & exchanging goods and services over computer network or internet through which transactions or items of sale are performed electronically.

: Benefits of E-Commerce

- Increased accessibility and convenience for customers
- Lower overhead costs compared to traditional brick-and-mortar stores
- Ability to reach a global market and target specific demographics

- Using online catalogs
  - Good online stores have a catalog
  - Products are often arranged by category
  - Products can be found by keyword



Process of E-commerce

Features of E-commerce:

- **E-Commerce provides following features**
- **Non-Cash Payment:** E-Commerce enables use of credit cards, debit cards, smart cards, electronic fund transfer via bank's website and other modes of electronics payment.
- **24x7 Service availability:** E-commerce automates business of enterprises and services provided by them to customers are available anytime, anywhere. Here 24x7 refers to 24 hours of each seven days of a week.
- **Advertising :** E-commerce increases the reach of advertising of products and services of businesses.
- **Marketing :**It helps in better marketing management of products / services.

- **Improved Sales:** Using E-Commerce, orders for the products can be generated any time, any where without any human intervention. By this way, dependencies to buy a product reduce at large and sales increases.

- **Support:** E-Commerce provides various ways to provide pre sales and post sales assistance to provide better services to customers.

- **Inventory Management:** Using E-Commerce, inventory management of products becomes automated. Reports get generated instantly when required. Product inventory management becomes very efficient and easy to maintain.

- **Communication improvement:** E-Commerce provides ways for faster, efficient, reliable communication with customers and partners.

Different types of e-commerce

I. Business-to-business (B2B)
II. Business-to-Consumer (B2C)
III. Business-to-government (B2G)
IV. Consumer-to-consumer (C2C)
V. Government to consumer (G2C)
VI. Government-to-business (G2B)

Digital Marketing Strategies

- Utilize social media platforms for brand promotion and customer engagement
- Implement SEO (Search Engine Optimization) techniques to improve online visibility
- Create compelling content such as blogs, videos, and infographics to attract and retain customers

Data Privacy and Security

- Protect customer data by complying with regulations such as GDPR (General Data Protection Regulation).
- Use encryption methods to secure online transactions and sensitive information.
- Regularly update security measures to prevent cyber threats.

Managing Online Reviews and Reputation

- Monitor online reviews and feedback from customers.
- Respond promptly to both positive and negative reviews in a professional manner.
- Build trust and credibility by maintaining a positive online reputation

Staying Updated and Adapting to Changes

- Stay informed about evolving trends and technologies in the online business landscape
- Be prepared to adapt strategies and embrace innovation to stay ahead of the competition
- Continuously learn and improve to meet the changing needs of your target audience

- Paying for purchases
  - Vendors accept several payment methods
    - Credit cards
    - Paypal
    - Electronic Fund Transfer
  - May need to create an account
    - Simplifies shipping
    - Encourages repeat shopping